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# Lab 1: Student Performance Analysis
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# 1. Create Student Data
# Variables and vectors
names <- c("Amaka", "Bayo", "Basse", "David", "Luke")
scores <- c(85, 67, 90, 72, 88)
gender <- c("F", "M", "M", "M", "F")

# Factor
gender <- factor(gender)

# Data frame
students <- data.frame(names, scores, gender)
print(students)

# =====
# 2. Grade Classification (Loops & Conditions)
# =====
grades <- c()

for (i in scores) {
  if (i >= 85) {
    grades <- c(grades, "A")
  } else if (i >= 70) {
    grades <- c(grades, "B")
  } else {
    grades <- c(grades, "C")
  }
}

students$grades <- grades
print(students)

# =====
# 3. Summary Statistics
# =====
mean_score <- mean(scores)
max_score <- max(scores)
min_score <- min(scores)

print(mean_score)
print(max_score)

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print(min_score)

# =====
# 4. Visualization
# =====

# Bar plot
barplot(scores, names.arg = names, col = "skyblue",
        main = "Student Scores")

# Histogram
hist(scores, col = "green", main = "Score Distribution")
```